

Yuhao Ge

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Education

University of Illinois at Urbana-Champaign	Master of Computer Science	2023.8 – 2025.1
University of Illinois at Urbana-Champaign	B.Sc. Computer Engineering GPA: 4.0/4.0	2019.8 - 2023.5
■ Honors: Highest Honors, Bronze Tablet (3%, 2023), Dean's List (2020&2022)		
Zhejiang University	B.Eng. Computer Engineering GPA: 3.93/4.0	2019.8 - 2023.5
■ Honors: Zhejiang Provincial Government Scholarship, First-Class Scholarship (3%, 2020&2022)		

Skills

- **Programming:** Python, C/C++, JavaScript, SystemVerilog, Assembly, SQL, MongoDB, Neo4j
- **Frameworks & Tools:** CUDA, PyTorch, TensorFlow, XLA, TVM, Triton, PyG, Flask, React, Node.js, Kubernetes, AWS, GCP

Work Experience

Amazon, Annapurna Labs <i>Software Engineer Compiler, Systems, ML, Accelerators</i>	2025.3 – now
■ Neuron Compiler <i>Optimize deep learning on AWS AI accelerators (Trainium and Inferentia)</i>	
• Developed a loop software pipelining pass supporting user-guided scheduling and deterministic transformations.	
• Implemented modulo scheduling with a cost model for compiler-driven optimization, and added autotuning for continuous performance improvement.	
• Designed and extended the Neuron Kernel Interface (NKI) to provide fine-grained user control and improved usability.	
• Integrated user-configurable backend passes (e.g., vectorization, loop scheduling) into the kernel compilation flow	
Amazon, Annapurna Labs <i>Software Engineer Intern Compiler, Systems, ML, Accelerators</i>	2024.5 – 2024.8
■ Neuron Compiler <i>Optimize deep learning on AWS AI accelerators (Trainium and Inferentia)</i>	
• Delivered three organization-wide presentations and received a Certificate of Appreciation for developing one of the most influential internship projects, which significantly enhanced performance on Trainium and opened up a new direction for future development.	
• Developed infrastructure for automatic kernel generation, compilation, profiling, and visualization with a defined sweep space	
• Collected data for DMA access pattern analysis and introduced a learning-based DMA latency model	
• Created the first-generation Autotuning infrastructure from scratch for compiler optimization and kernel optimization	
• Used autotuning to optimize the Matrix Multiply Fusion Pass , achieving a 14.7% improvement for the Llama3.1 model	
• Developed kernel language for AI accelerators and supported the kernel optimization with autotuning, achieving a 4.9% HFU improvement for kernels like Matrix Multiply	
• Implemented multi-process compilation and distributed benchmarking , resulting in 8.62X speedup	
TikTok <i>Software Engineer Intern C++, Lua, Game Engine, AR/VR</i>	2022.5 - 2022.8
■ Amazing Engine <i>TikTok's Next-Generation 3D Game Engine for AR/VR Effects</i>	
• Collaborated in developing TikTok's 3D Game Engine , which empowers users to create/use interactive AR/VR stickers	
• Implement a query-based animation system Motion Matching in C++ for realistic and responsive avatar control	
• Developed an SDK for Skeleton Retargeting in C++ and Lua , supporting animation adaptation across character models	
• Integrated the cross-functional team's Text-to-Animation algorithm into our game engine using the developed SDK	
University of California, Los Angeles <i>Visiting Student Researcher ML, RL, GNN, FPGA, EDA</i>	2022.6 - 2022.11
■ GNNDSE <i>An automated design space exploration for automatic FPGA accelerator design Advisor: Prof. Jason Cong</i>	
• Combined GNN with an ML/RL-based Design Space Exploration to achieve FPGA Accelerator Design Automation	
• Developed a learning-based Cost Model with GNN as a surrogate of the HLS tool for quick and accurate assessment	
• Optimized DSE by deploying heuristic algorithms such as Genetic Algorithm and Simulated Annealing	
• Used Reinforcement-Learning and Bandits for automatic algorithm selection, boosting exploration speed by 11%	

Selected Projects

Optimized GPU code generation framework for SParse reguLar Attention	2023.8 – 2024.7
• Developed SPLAT, an optimized framework for efficient sparse-MHSA , targeting moderate sparsity levels	
• Introduced Affine Compressed Sparse-Row (ACSR) format for regular sparsity patterns in MHSA	
• Engineered advanced GPU code-generation algorithms for ACSR, enhancing sparse-MHSA kernel performance	
• Achieved 2.05x and 4.05x speedups over Triton and TVM kernels with SPLAT implementation	
GoPricing: An NFT pricing service powered by machine learning	2023.2 – 2023.6
• Developed a Regression Model for NFT pricing using historical transaction data and NFT features	
• Design a system to support online inference and offline training , testing, and maintaining	
• Implemented with FastAPI , Airflow , MongoDB , and Redis , deployed with Kubernetes , and monitored through Grafana	
Remote Car Control System with Real-time 3D Reconstruction	2023.1 – 2023.5
• Developed a Raspberry Pi robot car with remote control via joysticks, utilizing PID control and STM32 microcontroller	
• Implemented WiFi-based communication for transmission of commands and RGBD images between the car and server	
• Implemented the SLAM framework RTAB-Map on the server for real-time 3D reconstruction , achieving a 10Hz framerate	

Implement A [Game](#) Efficiently on the FPGA Board

2022.1 - 2022.5

- Ported the game "Doodle Dump" to FPGA with **SystemVerilog**, achieving low power consumption and high efficiency
- Implemented a SOC with NIOS II in C to manage complex tasks like USB protocol and memory I/O
- Consumed only 400KB memory, 0.5w power to achieve a 50hz frame rate, won the **Best Design Prize**

Selected Research

- [SPLAT](#): A framework for optimised GPU code-generation for SParse reguLar ATtention